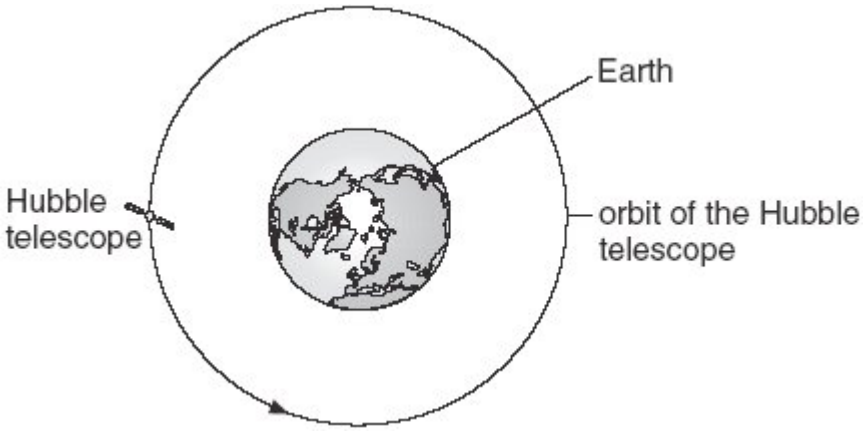
1. The diagram below shows the Hubble telescope in orbit around the Earth.



not to scale

(a) Which force keeps the telescope in orbit around the Earth?
Tick the correct box.

air resistance	☐	friction	☐
gravity	☐	magnetism	☐

1 mark

(b) The Hubble telescope is a satellite used for looking at planets and stars.

Give **one** other use of satellites.

.....

.....

1 mark

(c) Fill each of the gaps in the following sentences with a different word from the box below.

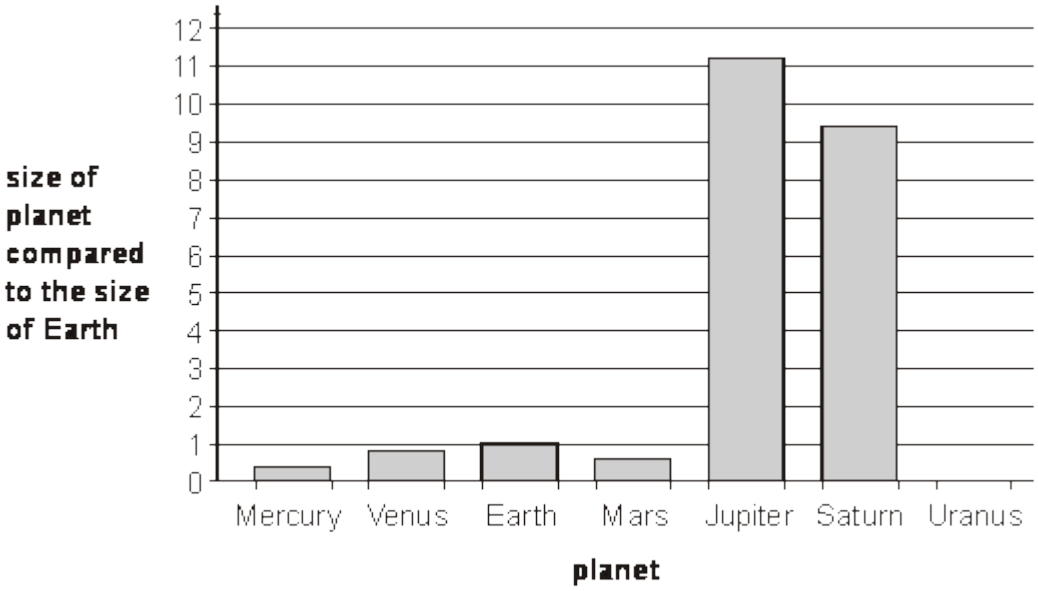
absorbs	**produces**	**reflects**

You can see the Sun because it ..... light.

You can see a satellite because it ..... light.

1 mark

(d) The bar chart shows the size of five planets compared to the size of Earth.



The planet Uranus is four times the size of Earth.
On the chart above, draw a bar for the planet Uranus.

1 mark

(e) (i) Arrange the following in order of size, starting with the smallest.

Sun	**Hubble telescope**	**Earth**

....
smallest **largest**

1 mark

(ii) Some stars are bigger than the Sun but they look smaller.
Why do they look smaller than the Sun?
Tick the correct box.

They are brighter than the Sun.	☐	They are the same colour as the Sun.	☐
They are further away than the Sun.	☐	They are nearer than the Sun.	☐

1 mark
maximum 6 marks

2. Our solar system consists of planets orbiting the Sun. Some of these planets have moons.

(a) Complete the four statements by drawing lines to join the boxes.

the average distance from Mercury to the Sun is	150 million km
the average distance from Pluto to the Sun is	0.38 million km
the average distance from the Earth to the Moon is	5900 million km
the average distance from the Earth to the Sun is	58 million km

4 marks

(b) Give the name of the star in our solar system.

.....

1 mark

Maximum 5 marks

1.

(a) • gravity ✓

if more than one box is ticked, award no mark

1 (L4)

(b) any **one** from

• telecommunications

accept 'communication'

• television **or** TV

• telephones

*accept 'phone' **or** 'fax'*

• radio

• weather

• military **or** spy observations

• land use observations

• GPS

accept 'navigation'

• internet

• monitoring **or** taking pictures of the Earth

*'looking at **or** taking pictures of planets **or** stars **or** galaxies' is insufficient as it is given in the question*

1 (L4)

(c) • produces
reflects

answers must be in the correct order
***both** answers are required for the mark*

1 (L4)

(d) • a bar drawn to 4

1 (L4)

(e) (i) • Hubble telescope Earth Sun

*accept 'Hubble' **or** 'telescope' for Hubble telescope*
answers must be in the correct order
all three answers are required for the mark

1 (L3)

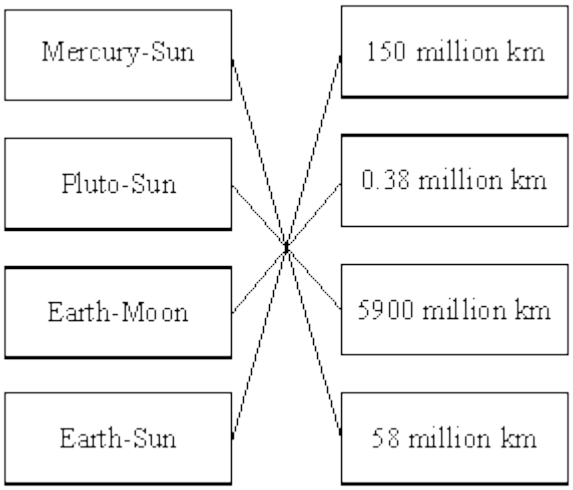
(ii) • They are further away than the Sun ✓

if more than one box is ticked, award no mark

1 (L3)

[6]

2. (a)



award a mark for each correct line

if more than four lines are drawn deduct one mark for each extra line

4

(b) the Sun

1

[5]